MILITARY SPECIFICATION SHEET

HOSE ASSEMBLIES, BREATHING OXYGEN, HIGH PRESSURE, WITH CONNECTIONS FOR AN INTEGRATED COMMUNICATION SYSTEM

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the hose assembly described herein shall consist of this specification and the latest issue of MIL-H-81581.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for high pressure oxygen hose assemblies, with connections for integrated communication systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-H-81581	-	Hose Assemblies, Breathing Oxygen and Air; General Specification for.

2.1.2 Other Government documents, drawings, and publications. The following other Government document forms a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

NAVAL AIR SYSTEMS COMMAND

278AS100 - Hose Assembly Index, Breathing Oxygen, High Pressure with Connections for an Integrated Communication System.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with the general specification MIL-H-81581 and as specified herein.

3.2 Qualification. The hose furnished under this specification shall be a product which is qualified for listing on the applicable Qualified Products List at the time set for opening of bids. Qualification will be in accordance with groups as listed on NASC Index Drawing 278AS100. Qualification approval of any individual hose listed within a group will automatically qualify a manufacturer for all other items listed within the group provided the conditions set forth in MIL-H-81581, paragraph 6.3.1, are complied with.

3.3 Talc. No talc shall be used in the manufacturing process of the hoses furnished under this specification.

3.4 Design and construction. The hose assembly shall be in accordance with the general specification and with the applicable drawing listed on NASC Index Drawing 278AS100.

3.5 Performance. Unless otherwise specified herein, the performance of the hose assembly shall be as specified in the general specification, consistent with the applicable performance requirements for the applicable tests and inspections specified herein.

3.5.1 Polarity (color code) and electrical continuity. When tested as specified in the general specification, the resistance measured between all points designed to be electrically connected shall be as specified in the general specification with an allowable resistance of 0.075 ohms per foot length of the electrically wired portion of the hose.

3.5.2 Elongation.

3.5.2.1 Procedure I. When tested as specified in the general specification, the increase in length of the hose shall not exceed 12 percent of the original hose length.

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3.5.2.2 Procedure II. When tested as specified in the general specification, the permanent set of the hose shall not exceed 2 percent of the original hose length when measured 2 minutes after removal of load.

3.5.3 Strength.

3.5.3.1 Tensile load. When tested as specified in the general specification, there shall be no indication of separation of the end fittings or molded ends from the hose assembly.

3.5.3.2 Static load. When tested as specified in the general specification, the hose shall not show a deflection of more than 25 percent of original value on the outside diameter. With the load removed, the outside diameter of the hose shall be within 5 percent of original value.

3.5.4 Restraint cord elongation and tensile strength. When the applicable control drawing specifies a restraint cord, the restraint cord shall have a 30 percent maximum elongation and a minimum tensile strength of 70 pounds when tested as specified in the general specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with the general specification and as specified herein.

4.1.1 Qualification inspection. The qualification inspection of the hose assembly shall consist of the following examinations and tests from the general specification:

Visual examination
Polarity (color code) and electrical continuity, procedure I
Insulation resistance, procedure I
Elongation, procedure II
Odor, procedure I
Cleanliness, procedures I and II
Leakage, procedure III
Flexibility, procedure IV (under environmental conditions procedure II)
Minimum burst pressure, procedure III
Tensile load, procedure II and procedure III (if applicable)
Static load, procedure I
Restraint cord elongation and tensile strength
Low temperature, procedure I
High temperature, procedure I
Flexibility endurance, procedure II
Ozone-resistance, procedure I
Abrasion, procedure I

4.1.2 First article inspection. The first article inspection of the hose assembly shall consist of the applicable examinations and tests specified in the general specification with the applicable procedure as specified herein.

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4.1.2.1 First article samples. The first article samples shall consist of four hose assemblies of the part number contracted for.

4.1.3 Quality conformance inspection. The quality conformance inspection shall consist of the following tests from the general specification. The sample size and acceptance criteria shall be as specified in the general specification.

Visual examination
Polarity (color code) and electrical continuity, procedure I
Insulation resistance, procedure I
Elongation, procedure II
Odor, procedure I
Cleanliness, procedures I and II
Leakage, procedure III
Flexibility, procedure IV
Packaging

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the general specification.

6. NOTES

6.1 Intended use. The hose assemblies covered by this detail specification are intended for use in breathing oxygen, and contain integrated communication services for aircrewmen in the operation of the aircraft.

Preparing Activity:
Navy - AS
(Project No. 1660-N445)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-H-81581/2A(AS) Hose Assy, Breathing Oxygen, High Pressure, with Connections for an Integrated Communication System

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

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DD FORM 1 OCT 76 1426 Replaces edition of 1 Jan 72 which may be used. S/N 0102-LF-001-4260

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